

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - X

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- a. Create a PWA with a Web App Manifest file including: App name Icons Start URL Display mode Implement Push Notifications in a PWA.**

Manifest.json:

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

Html

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <h1>My Progressive Web App</h1>
  <button onclick="showNotification()">
    Show Notification
```

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```
</button>
<script src="app.js"></script>
</body>
</html>
```

app.js:

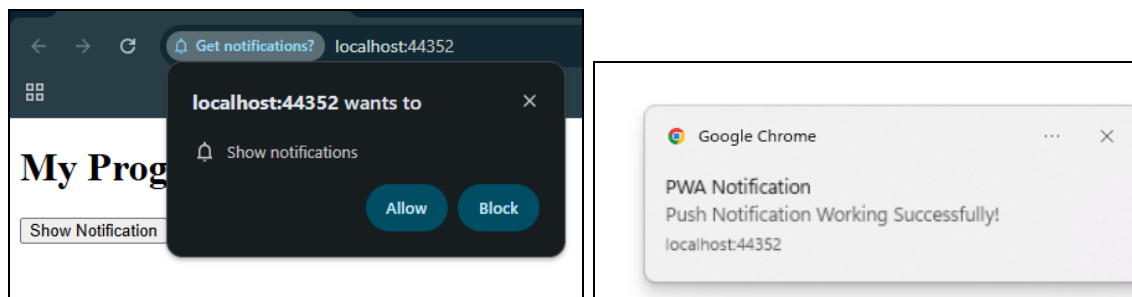
```
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}

function showNotification() {
  Notification.requestPermission().then(permission => {
    if (permission === 'granted') {
      navigator.serviceWorker.ready.then(registration => {
        registration.showNotification('PWA Notification', {
          body: 'Push Notification Working Successfully!',
          icon: 'icons/icon1.png'
        });
      });
    }
  });
}
```

Service-wroker:

```
self.addEventListener('install', event => {
  console.log('Service Worker Installed');
});

self.addEventListener('activate', event => {
  console.log('Service Worker Activated');
});
```



b. Develop an E-Commerce PWA with product listing and offline cart functionality.

Program.cs

```
var builder = WebApplication.CreateBuilder(args);  
var app = builder.Build();  
app.UseDefaultFiles();  
app.UseStaticFiles();  
app.Run();
```

manifest.json

```
{  
  "name": "E-Commerce PWA",  
  "short_name": "ShopPWA",  
  "start_url": "/index.html",  
  "display": "standalone",  
  "background_color": "#ffffff",  
  "theme_color": "#d6d6d6",  
  "icons": [  
    {  
      "src": "icons/icon1.png",  
      "sizes": "192x192",  
      "type": "image/png"  
    },  
    {  
      "src": "icons/icon2.png",  
      "sizes": "512x512",  
      "type": "image/png"  
    }  
  ]  
}
```

index.html

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>E-Commerce PWA</title>  
  <link rel="manifest" href="manifest.json">  
  <meta name="theme-color" content="#d6d6d6">  
<style>  
  body {  
    font-family: Arial;  
    margin: 20px;  
  }  
  .container {  
    display: flex;  
    gap: 30px;  
  }  
  .cart-section {  
    width: 30%;  
    border: 1px solid gray;  
    padding: 15px;
```

```
        min-height: 300px;
    }
    .product-section {
        width: 70%;
    }
    .products {
        display: flex;
        flex-wrap: wrap;
        gap: 15px;
    }
    .product {
        border: 1px solid gray;
        padding: 10px;
        width: 200px;
    }
    button {
        padding: 5px 10px;
        cursor: pointer;
    }
</style>
</head>
<body>
    <h1>E-Commerce PWA</h1>
    <div class="container">
        <!-- Cart Section -->
        <div class="cart-section">
            <h2>Cart Items</h2>
            <ul id="cartList"></ul>
        </div>
        <!-- Product Section -->
        <div class="product-section">
            <h2>Products</h2>
            <div class="products">
                <div class="product">
                    <h3>Laptop</h3>
                    <p>Price: ₹ 50000</p>
                    <button onclick="addToCart('Laptop')">
                        Add to Cart
                    </button>
                </div>
                <div class="product">
                    <h3>Mobile</h3>
                    <p>Price: ₹ 20000</p>
                    <button onclick="addToCart('Mobile')">
                        Add to Cart
                    </button>
                </div>
                <div class="product">
                    <h3>Headphones</h3>
```

```
        <p>Price: ₹ 3000</p>
        <button onclick="addToCart('Headphones')">
            Add to Cart
        </button>
    </div>
</div>
</div>
</div>
<script src="app.js"></script>
</body>
</html>
```

app.js

```
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
function addToCart(product) {
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart));
}
```

```
function displayCart() {
    let cartList = document.getElementById('cartList');
    cartList.innerHTML = "";
    cart.forEach(item => {
        let li = document.createElement('li');
        li.textContent = item;
        cartList.appendChild(li);
    });
}
```

service-worker.js

```
const CACHE_NAME = 'ecommerce-cache-v1';
const urlsToCache = [
    '/',
    '/index.html',
    '/app.js',
    '/manifest.json'
];
self.addEventListener('install', event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});
self.addEventListener('fetch', event => {
    event.respondWith(
```

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```
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
    );
  });
```

